

Soil Health & Crop Recommendation Report

Report Date: November 20, 2025 at 22:49:17 UTC

Farmer Information	
Name:	ganesh
Phone:	8088117817
Location:	vijayapura
Land Size:	1.00 acres

Parameter	Value	Unit
Nitrogen (N)	50.0	mg/kg
Phosphorus (P)	50.0	mg/kg
Potassium (K)	50.0	mg/kg
pH Level	6.50	
Temperature	25.0	°C
Humidity	60.0	%
Rainfall	120.0	mm
Soil Type	Black Soil	

Soil Health Assessment

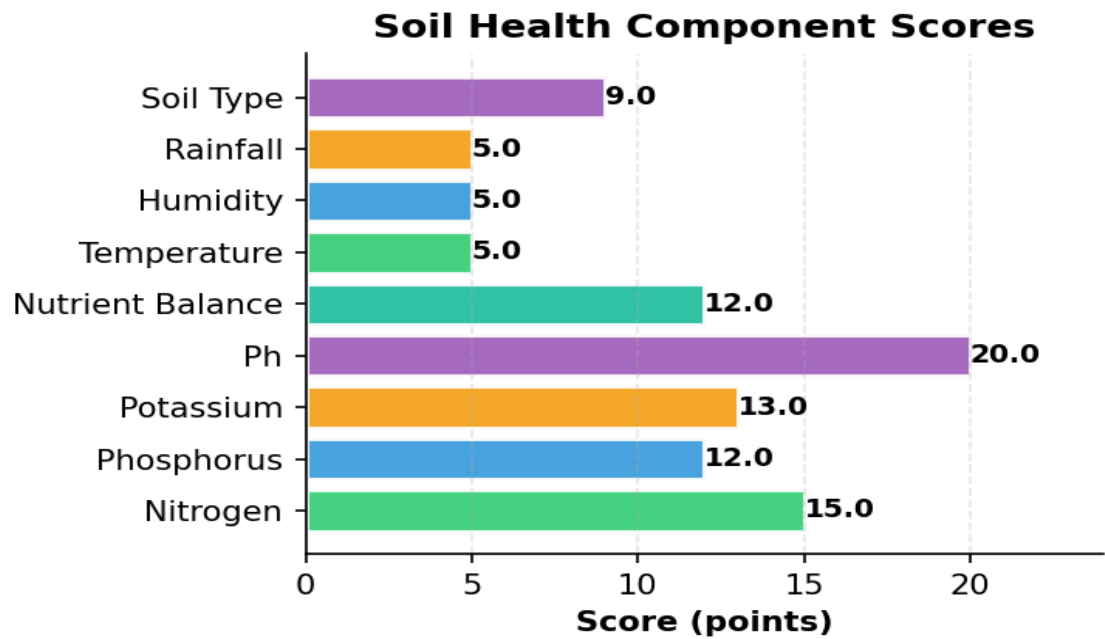
Overall Soil Health Score	96.0/100	Excellent	
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Component	Score	Points
Nitrogen	15.0	points
Phosphorus	12.0	points
Potassium	13.0	points
Ph	20.0	points
Nutrient Balance	12.0	points
Temperature	5.0	points
Humidity	5.0	points
Rainfall	5.0	points

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Soil Type	9.0	points
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Weather Suitability for Pigeonpeas

Overall Suitability Score: 7.3/10 Good

Weather conditions are good for this crop

- ✓ Temperature 25.0°C is ideal for pigeonpeas
- ✓ Humidity 60.0% is ideal for pigeonpeas — reduces disease risk
- Rainfall 120 mm is below requirement (600-1000 mm); irrigation strongly recommended

Recommended Crop

Predicted Crop	Pigeonpeas
Confidence Level	82.9%
Model Used	Stacking Classifier

Top 3 Alternative Crop Options

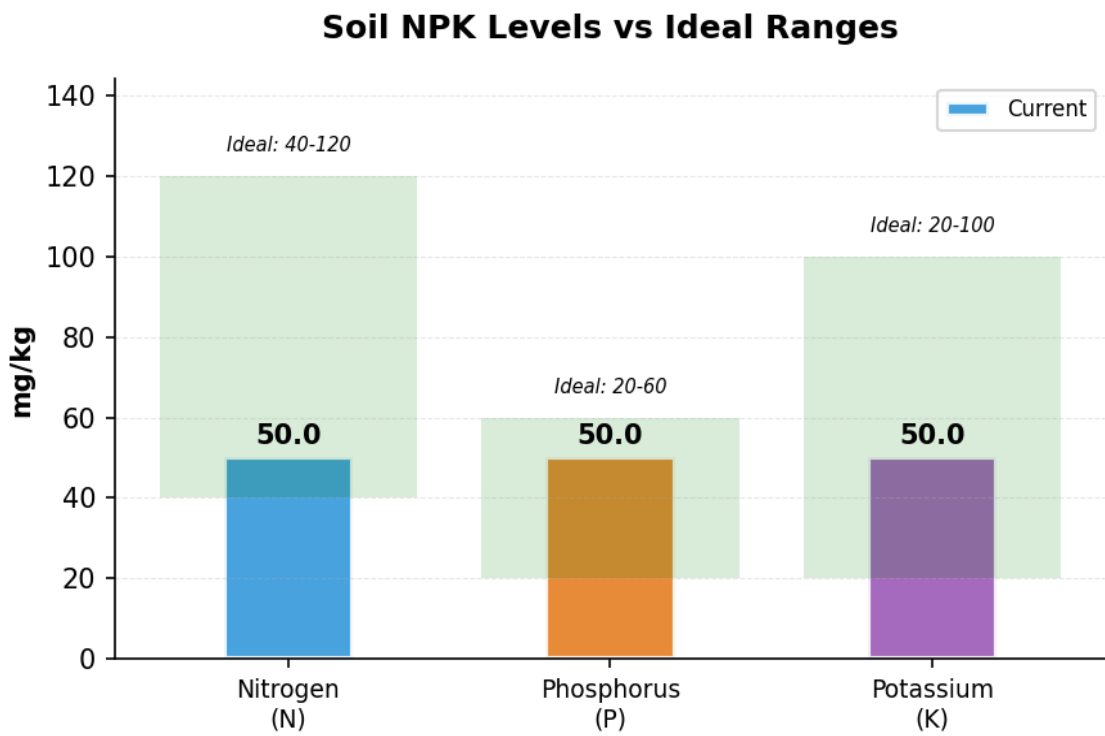
Rank	Crop	Confidence (%)
1	Jute	3.2%

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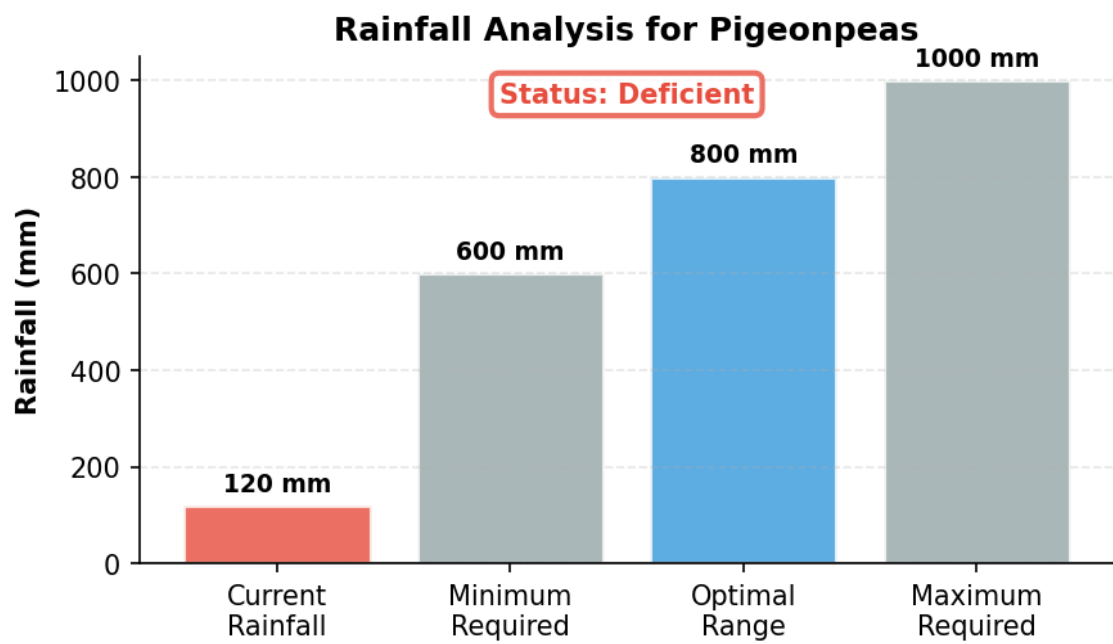
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2	Rice	1.7%
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Soil NPK Levels Visualization



Rainfall Analysis



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■ Warnings & Observations

Action Required:

- Rainfall below optimal → Increase irrigation

Positive Indicators:

- ✓ Nitrogen level optimal
- ✓ Phosphorus level optimal
- ✓ Potassium level optimal
- ✓ pH optimal (6.5)
- ✓ Soil type: Black Soil

■ Market Price & Profitability Analysis

Price Range	2000 - 3500 ■/Quintal
Average Price	2750 ■/Quintal

Best Market: Local Market, Your State (2750 ■/Quintal)

Market Advice: Market data integration is available in the system. In a production environment, this would show real-time market prices.

■ Soil Treatment Recommendations

- Soil pH (6.5) is within optimal range (6.0-7.5) for pigeonpeas.
- Nitrogen (50.0 mg/kg) is excessive for pigeonpeas. Optimal range: 10-30. Reduce nitrogen application to avoid excessive vegetative growth.
- Phosphorus (50.0 mg/kg) is within optimal range for pigeonpeas.
- Potassium (50.0 mg/kg) is excessive for pigeonpeas. Optimal range: 10-30. High K can cause magnesium deficiency.
- High clay content, retains moisture well. Ideal for cotton, wheat, and pulses.

■ Fertilizer Recommendations

FARM DETAILS

Land Size: 1.0 acres
Current Soil NPK: N=50.0 | P=50.0 | K=50.0 mg/kg
Optimal NPK for pigeonpeas: N=20 | P=50 | K=20 kg/acre

SOIL NUTRIENT STATUS

- ✓ Excellent: Soil NPK levels are already optimal for pigeonpeas

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Maintain soil health through:

- Regular organic compost application
- Crop rotation practices
- Green manuring between seasons

ORGANIC & BIO-FERTILIZER ALTERNATIVES

Farm Yard Manure (FYM): 5-10 tons

- ■ Application: 5-10 tons/acre (provides balanced NPK)

Compost: Well-decomposed organic compost

- ■ Application: 3-5 tons total

Green Manure: Dhaincha, Sunhemp, or other leguminous crops

- ■ Benefits: Natural nitrogen fixation, improves soil structure

Bio-Fertilizers:

- Rhizobium: 200g/acre (for nitrogen fixation)
- PSB (Phosphate Solubilizing Bacteria): 200g/acre
- Azospirillum: 200g/acre (for nitrogen fixation)
- VAM (Vesicular Arbuscular Mycorrhiza): Enhances P uptake

Note: Apply fertilizers based on soil test results and local agricultural extension recommendations for best results.

■ Crop-Specific Tips for Pigeonpeas

Drought-tolerant legume. Requires well-drained soil and warm climate.

■ Additional Notes & Disclaimer

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.